Knowledge and Data

Working Group

Week 2: RDF & SPARQL

VRIJE
UNIVERSITEIT
AMSTERDAM

Question 1—Modelling tabular information as a graph

Consider the following *relational database* tables:

Students

student	*VUnetID*	*uni*	*study*
s1	as344	UvA	IK
s2	ex444	VU	IMM

StudyProgrammes

study	*name*	*language*
IMM	Information, Management, and Multimedia	Dutch
IK	Informatiekunde	Dutch

Task: Construct a knowledge graph G that represents this knowledge, using our simple triple notation:

$$(\textit{subject}, \textit{predicate}, \textit{object})$$

Question 2—RDF and Turtle

Consider once more our knowledge graph G.

Task: Serialize the triples that make up G in *Turtle* syntax.
Such that you:

1. make use of namespaces and URIs / IRIs,
2. abbreviate namespaces using prefixes, and
3. use one or more syntactic shortcuts.

Question 2—RDF and Turtle

The entire graph:

(s1,	a,	Student)
(s2,	a,	Student)
(s1,	hasVUnetID,	"as344")
(s2,	hasVUnetID,	"ex444")
(s1,	studiesAt,	UvA)
(s2,	studiesAt,	VU)
(s1,	followsStudy,	IK)
(s2,	followsStudy,	IMM)
(IMM,	a,	StudyProgramme)
(IK,	a,	StudyProgramme)
(IMM,	hasName,	"Information, [...]")
(IK,	hasName,	"Informatiekunde")
(IMM,	hasLanguage,	Dutch)
(IK,	hasLanguage,	Dutch)

Turtle specification: https://www.w3.org/TR/turtle

Question 3—Models for knowledge graphs

Consider a knowledge graph H with triples (a, p, b) and (b, p, c) :

$$a \xrightarrow{p} b \xrightarrow{p} c$$

Let an interpretation I consist of universe $U = \{1, 2\}$
Which of the following interpretations is a model of H:

1. $I(a) = 2, I(b) = 1, I(c) = 1, I(p) = \{(1, 2)\}$
2. $I(a) = 2, I(b) = 1, I(c) = 2, I(p) = \{(1, 2)\}$
3. $I(a) = 2, I(b) = 1, I(c) = 2, I(p) = \{(2, 2)\}$
4. $I(a) = 2, I(b) = 1, I(c) = 1, I(p) = \{(2, 2), (1, 1)\}$
5. $I(a) = 2, I(b) = 2, I(c) = 2, I(p) = \{(2, 2)\}$

Remember: I is a model of a knowledge graph iff it is a model of *all* its triples, and I is a model of a triple iff the interpretation is true

Question 4—Entailments for knowledge graphs

Let K be a knowledge graph defined by triples

(Victor,	teaches,	KaD)
(Stefan,	a,	Teacher)
(KaD,	a,	Course)
(Carl,	participatesIn,	KaD)
(Carl,	a,	Student)

Which of the following knowledge graphs is entailed by K?

1) (Carl, participatesIn, Course)

2) (Victor, teaches, Course)
(Stefan, a, Teacher)
(KaD, a, Course)
(Carl, participatesIn, Course)

3) (Stefan, a, Teacher)
(Carl, a, Student)
(KaD, a, Course)

4) (Victor, teaches, KaD)
(Stefan, a, Teacher)
(KaD, a, Course)
(Carl, participatesIn, KaD)
(Carl, a, Student)
(Carl, participatesIn, Course)

Question 5—Resources

Describe, in your own words, the difference between a URI/IRI and a Literal

Question 6—Exploring DBPedia

Consider the following two DBpedia resources:

a) https://dbpedia.org/resource/Amsterdam

b) https://dbpedia.org/resource/Kingdom_of_the_Netherlands

Which of the follow three triples together form a path from **a** to **b**:

1. (dbr:Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
2. (dbr:Kingdom_of_the_Netherlands, dbo:language, dbr:Dutch_language)
3. (dbr:Amsterdam, rdf:type, dbo:Place)
4. (dbr:Kingdom_of_the_Netherlands, dbo:leaderName, dbr:Willem-Alexander_of_the_Netherlands)
5. (dbr:Netherlands, dbo:part_of, dbr:Kingdom_of_the_Netherlands)
6. (dbr:Willem-Alexander_of_the_Netherlands, rdf:type, dbo:Person)
7. (dbr:Amsterdam, dbo:country, dbr:Netherlands)

Question 7—DBpedia & SPARQL

Consider following SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?capital ?area
WHERE {
   ?country dbo:capital ?capital.
   ?capital dbo:areaTotal ?area .
}
```

Execute this query against the DBpedia SPARQL endpoint [1] in YASGUI [2].

[1] http://dbpedia.org/sparql
[2] https://yasgui.triply.cc

Question 8—Eliminate duplicates with SPARQL

Consider, once again, the following SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT ?capital ?area
WHERE {
   ?country dbo:capital ?capital.
   ?capital dbo:areaTotal ?area .
}
```

Adjust the query to eliminate duplicate results. How many remain?

SPARQL specification: https://www.w3.org/TR/sparql11-query/

Question 9—SPARQL Advanced

Consider our improved SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
SELECT DISTINCT ?capital ?area
WHERE {
   ?country dbo:capital ?capital.
   ?capital dbo:areaTotal ?area .
}
```

Restrict the query to include only bindings to `?capital` that have a *rdfs:label* relation with the English language literals "Amsterdam" or "Berlin".

Note that there is more than one solution

Question 10—SPARQL Advanced II

Consider our filtered SPARQL query:

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX dbpedia: <http://dbpedia.org/resource/>
SELECT DISTINCT ?capital ?area
WHERE {
   ?country dbo:capital ?capital .
   ?capital dbo:areaTotal ?area ;
             rdfs:label ?label .

   FILTER (?label = "Amsterdam"@en || ?label = "Berlin"@en)
}
```

Extend the query to also retrieve the value of the *dbo:areaWater* property and bind it to a variable `?areaWater`.

Make sure that it still returns both Amsterdam and Berlin.

Quiz

To test your proficiency:

Exercise quiz Module 2

(see the course's Canvas page)

NB: completing the quiz is mandatory to continue to next week's module